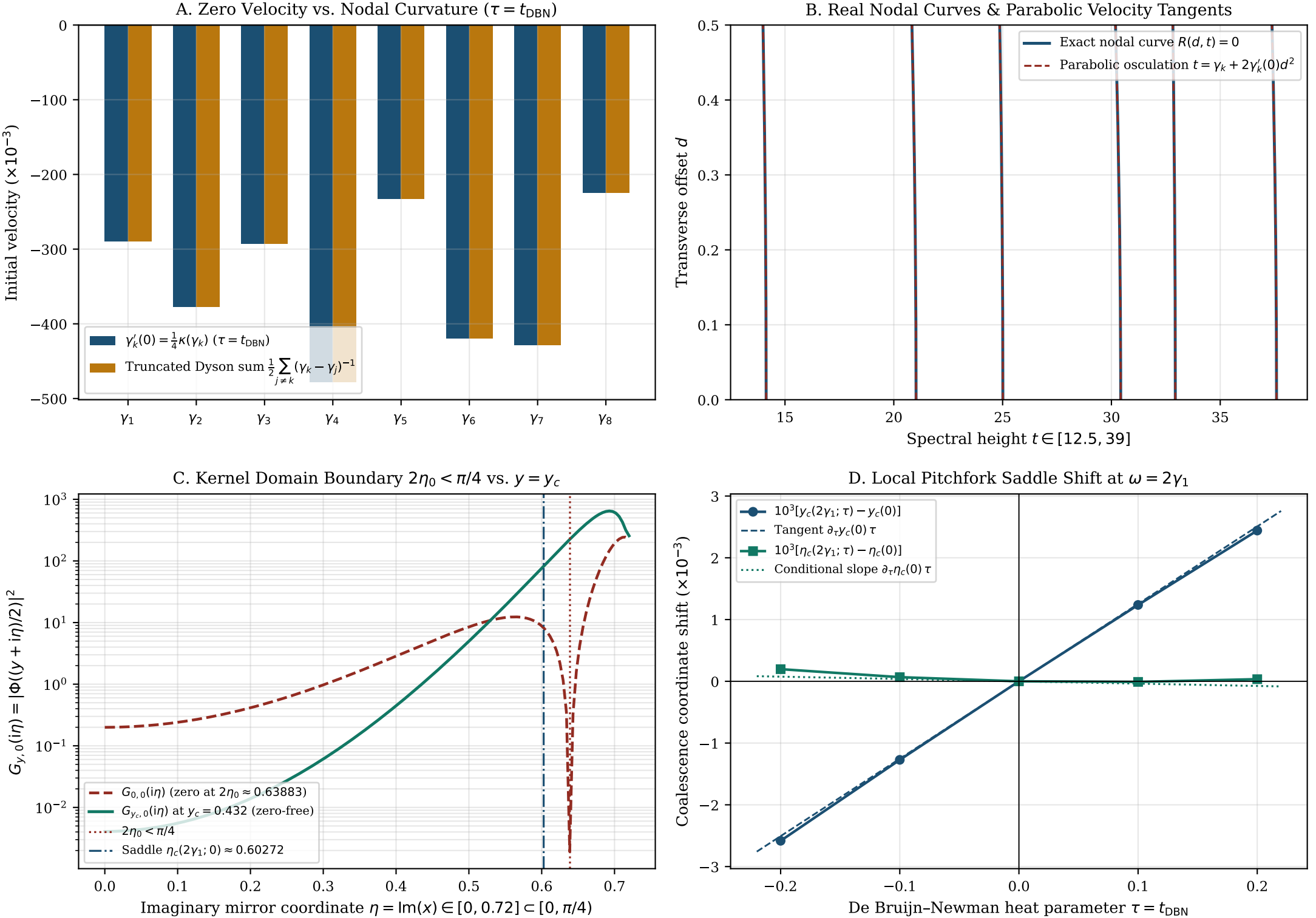

A. Zero Velocity vs. Nodal Curvature ($\tau = t_{\mathrm{DBN}}$)
Initial velocity ($\times 10^{-3}$)
0
−100
−200
−300
−400
−500
γ_1
γ_2
γ_3
γ_4
γ_5
γ_6
γ_7
γ_8
$\gamma_k'(0) = \frac{1}{4}\kappa(\gamma_k)$ ($\tau = t_{\mathrm{DBN}}$)
Truncated Dyson sum $\frac{1}{2}\sum_{j\neq k}(\gamma_k - \gamma_j)^{-1}$
B. Real Nodal Curves & Parabolic Velocity Tangents
Transverse offset d
0.5
0.4
0.3
0.2
0.1
0.0
15
20
25
30
35
Spectral height $t \in [12.5, 39]$
Exact nodal curve $R(d, t) = 0$
Parabolic osculation $t = \gamma_k + 2\gamma_k'(0)d^2$
C. Kernel Domain Boundary $2\eta_0 < \pi/4$ vs. $y = y_c$
$G_{y,0}(i\eta) = |\Phi((y + i\eta)/2)|^2$
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Imaginary mirror coordinate $\eta = \mathrm{Im}(x) \in [0, 0.72] \subset [0, \pi/4)$
$G_{0,0}(i\eta)$ (zero at $2\eta_0 \approx 0.63883$)
$G_{y_c,0}(i\eta)$ at $y_c = 0.432$ (zero-free)
$2\eta_0 < \pi/4$
Saddle $\eta_c(2\gamma_1; 0) \approx 0.60272$
D. Local Pitchfork Saddle Shift at $\omega = 2\gamma_1$
Coalescence coordinate shift ($\times 10^{-3}$)
3
2
1
0
−1
−2
−3
−0.2
−0.1
0.0
0.1
0.2
De Bruijn–Newman heat parameter $\tau = t_{\mathrm{DBN}}$
$10^3[y_c(2\gamma_1; \tau) - y_c(0)]$
Tangent $\partial_\tau y_c(0)\,\tau$
$10^3[\eta_c(2\gamma_1; \tau) - \eta_c(0)]$
Conditional slope $\partial_\tau \eta_c(0)\,\tau$